

Dr. Deeksha Pandey (PhD)

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Portfolio:

<https://sites.google.com/view/mkuactivitycorner/manish-kumars-lab/deekshas-corner?authuser=>

https://www.researchgate.net/profile/Deeksha_Pandey4

https://scholar.google.co.in/citations?hl=en&user=V13V4i8AAAAJ&view_op=list_works&sortby=pubdate

<https://www.linkedin.com/in/deeksha-pandey-b1a76788/>

Research Interest

My most of research interests lie in **analyzing the different factors that contribute to the development of bacterial antibiotic resistance in microbial pathogens**. Most of my current research applies machine learning, pattern recognition, gene ontology, and molecular modeling & docking protocols that are suitable to convert large amounts of genomics/proteomics/metagenomics data information into knowledge. Shortly, I am very interested in extending my experience and knowledge of machine learning and computational approaches that would help in my study of antibiotic resistance genes. Using extended knowledge, we could be able to develop some patterns that would be used as a surveillance approach at an early stage of infection in pathogenic microbes as well as facilitating preventive actions to counter the spread of new variants of resistant bacteria in medicinal and clinical environments. I also have an elementary idea of cancer biology and analyzing a mutational signature at microsatellite repeats at healthy derived cells.

Education

Ph.D. thesis title - “**Characterization and analysis of antimicrobial resistance using *in-silico* tools.**” from Department of Biophysics, University of Delhi, South Campus, India (August 2016 - March 2023) Degree Awarded (24 February 2024).

M.Tech (Bioinformatics) from Banasthali University Banasthali, Rajasthan, India (2013-2015).

PG Diploma in Bioinformatics from Banasthali University Banasthali, Rajasthan, India (2012-2013).

B.Sc. Biotechnology from Dayanand Girls P.G. College, C.S.J.M. University, Kanpur, India (2009-2012).

Area of Specialization

Study of antibiotic resistance gene load and their evolution, Evolutionary and phylogenetic analysis of antimicrobial drug resistance, Classification of antimicrobial drug resistance proteins/genes, Basic analysis of Next Generation Sequencing (NGS) and Whole Genome Sequencing (WGS) data, Integration of -omics approaches, Compilation and development of web-resources (databases/web-tools), Management of large data sets (-omics) genomics/proteomics/metagenomics, Genome Alignment Tools, Data Mining, Machine Learning, Pattern Recognition & Database Development, Computational Biology & Molecular Modeling: 3-D Structure Prediction, Molecular Docking, Molecular Dynamics Simulations, Proteomics Data Analysis, Computer Aided Drug Discovery and Designing (CADD), Chem-informatics, Analysis of environmental metagenomics and human microbiome data, bio-algorithm development, Antimicrobial resistance databases and tools handling, preparing figures and statistical analysis using R packages.

Publications (Cumulative Impact Factor: 46.008, First Author Papers: 05)

1. **D Pandey**, I Gupta, D Gupta. AmpC β -lactamases: A Key to Antibiotic Resistance in ESKAPE Pathogens (doi: <https://doi.org/10.1101/2025.03.05.641667>)
2. D Mandal, **D Pandey**, DP Sarkar, M Kumar. Nucleoside and non-nucleoside reverse transcriptase inhibitor drugs (NRTIs and NNRTIs) are capable of binding Chandipura virus polymerase protein (L) and inhibiting virus replication. *VirusDisease* 2024 **PMID: 39464733 (IF: 1.848)**
3. **Pandey D**, Singhal N, Kumar M. β -LacFamPred: An online tool for prediction and classification of β -lactamase class, subclass, and family. *Frontiers in Microbiology* 2022 **PMID: 36713195 (IF: 6.064)**
4. Akhtar N, Mannan MA, **Pandey D**, Sarkar A, Sharma H, Kumar M, Ghosh A. Potent antifungal properties of gallic acid in *Sarcochlamys pulcherrima* against *Candida auris*. *BioTechnologia* (Pozn). 2023 Jun 26;104(2):105-119. **PMID: 37427031 (IF: 0.17)**
5. **Pandey D**, Kumari B, Singhal N, Kumar M. BacARscan: an *in silico* resource to discern diversity in antibiotic resistance genes. *Biology Methods and Protocols* 2022 Volume 7, Issue 1, 2022, bpac031 <https://doi.org/10.1093/biomethods/bpac031>. **PMID: 36479434 (IF: 3.6)**
6. **Pandey D**, Singhal N, Kumar M. Investigating the OXA variants of ESKAPE pathogens. *Antibiotics* 2021 Dec 15;10(12):1539. doi: 10.3390/antibiotics10121539. **PMID: 34943751 (IF: 4.9)**
7. **Pandey D**, Kumari B, Singhal N, Kumar M. BacEffluxPred: A two-tier system to predict and categorize bacterial efflux mediated antibiotic resistance proteins. *Scientific Reports* 2020 Jun 9; 10(1): 9287. doi: 10.1038/s41598-020-65981-3. **PMID: 32518231 (IF: 4.6)**
8. Singhal N, **Pandey D**, Singh NS, Kumar M, Viridi JS. Exploring the genetic determinants underlying the differential production of an inducible chromosomal cephalosporinase - BlaB in *Yersinia enterocolitica* biotypes 1A, 1B, 2, and 4. *Scientific Reports*. 2020 Jun 23; 10(1): 10167. doi: 10.1038/s41598-020-67174-4. **PMID: 32576927 (IF: 4.6)**

9. N Singhal, **D Pandey**, M Kumar, JS Viridi. Molecular analysis of ampR and ampD to understand variability in the inducible expression of “BlaB-like” cephalosporins in *Yersinia enterocolitica* biotype. 1A *Gene* 2019 704, 25-30. **PMID: 30980942 (IF: 3.913)**
10. N Singhal, **D Pandey**, NS Singh, M Kumar, JS Viridi. ampD homologs in biotypes of *Yersinia enterocolitica*: Implications in the regulation of chromosomal AmpC-type cephalosporinases. *Infection, Genetics and Evolution* 2019 69; 211-215. **PMID: 30710654 (IF: 4.393)**
11. N Singhal, **D Pandey**, NS Singh, M Kumar, JS Viridi. Molecular Characteristics of “BlaBlike” Chromosomal Inducible Cephalosporinase of *Yersinia enterocolitica* Biotype 1A Strains" for Microbial Drug Resistance. *Microbial Drug Resistance* 2019 DOI: 10.1089/mdr.2018.0282. **PMID: 30817214 (IF: 2.6)**
12. **Pandey D**, Podder A, Pandit M, Latha N, CD4-*gp120* Interaction Interface - A Gateway For HIV-1 Infection in Human: Molecular Network, Modeling and Docking Studies. *Journal of Biomolecular Structure and Dynamics*. 2016, 1-14. **PMID: 27545652 (IF: 5.235)**
13. Podder A, **Pandey D**, Latha N. Investigating the Structural Impact of S311C Mutation in DRD2 Receptor by Molecular Dynamics & Docking Studies. *Biochimie* 2016 123, 52-64. **PMID: 26805384 (IF: 4.079)**

Protocol

1. **Pandey D**, Kumari B, Singhal N, Kumar M. Protocol for detecting bacterial proteins involved in efflux-mediated antibiotic resistance (ARE) and their sub-families. *Nature Protocol Exchange* 2021 doi: <https://doi.org/10.21203/rs.3.pex-1371/v1>

Preprints

1. **Pandey D**, Kumari B, Singhal N, Kumar M. BacARscan: A Comprehensive and Interactive Web-Resource to Discern Antibiotic Resistance Gene Diversity in-Omics Datasets. 2021 doi: <https://doi.org/10.21203/rs.3.rs-712636/v1>
2. Dibyakanti Mandal, **Deeksha Pandey**, Debi P. Sarkar, Manish Kumar. Remdesivir, Zidovudine (AZT) and Nevirapine inhibit Chandipura virus replication through high-energy interactions with the RdRp domain of the polymerase protein L. doi: <https://doi.org/10.1101/2022.03.02.482698>

Web-resource/Standalone Tool

1. **BacEffluxPred**: *In-silico* two-tier prediction system for bacterial efflux conferring antibiotic resistance proteins and their subfamilies. <http://www.proteininformatics.org/mkumar/baceffluxpred/>
2. **BacARscan**: An in-silico resource to discern diversity in antibiotic resistance genes. <http://www.proteininformatics.org/mkumar/bacarscan/>
3. **β-LacFamPred**: HMM-based prediction and annotation tool for β-lactamase families <http://proteininformatics.org/mkumar/blacfampred/>

Professional Membership

1. Active member and contributor of Asia-Pacific Bioinformatics Network (**APBioNET**).
2. Member, BIOinformatics CLUb for Experimenting Scientists (**BIOCLUES**).
3. An active member of the Bioinformatics and Drug Discovery Society (**BIDDS**).
4. Member of the European Society of Clinical Microbiology and Infectious Diseases (**ESCMID**).
5. Global Outreach - Postdoctoral **American Society for Microbiology (ASM)** Member.
6. **Steering committee member** at “The 24th International Conference on Bioinformatics (InCoB 2025)” 18 – 20th Sept 2025 at the BOSE Institute, Kolkata, India.
(<https://incob.apbionet.org/incob2025/committee/>)

Editorial Roles

1. Editorial Board Member, **BMC Microbiology**, 2025–Present

Peer Review Contributions

Reviewed manuscripts for: **PeerJ, BMC Medical Genomics, Journal of Obstetrics and Gynaecology, Scientific Reports, Qeios Journal.**

Technical Skills

- ❖ Machine Learning & Pattern Recognition (PSSM, SVM, HMM)
- ❖ Computational Biology and Molecular Modeling (Protein-Protein Docking: Cluspro, Zdock, Patchdock, Hex, Gramm-X)
- ❖ Protein-Ligand Docking (Autodock, Schrodinger)
- ❖ Molecular structure prediction and visualization (I-TASSER, Rasmol, Jmol, Maestro, SPDBv, Pymol)
- ❖ Molecular Dynamics Simulation (Schrodinger Desmond)
- ❖ System Biology (CYTOSCAPE, String, Mint, Bind, DIP, HPRD Database)
- ❖ Gene Set Enrichment Analysis (DAVID database)
- ❖ Pathway Analysis (KEGG, Reactome databases)
- ❖ Molecular biology and biological databases (NCBI, Uniprot, EMBL, DDBJ, RCSB-PDB)
- ❖ Handling Bioinformatics Tools (Blast, ClustalW, Muscle, MEGA, Phylip, Procheck, ProtParam, Ligplot, Saves, X-score)
- ❖ Computer proficiency Linux, Windows, Mac
- ❖ Knowledge of various computer & Programming languages (PERL, HTML, CSS & R studio, Shell Scripting UNIX/LINUX)
- ❖ Knowledge of working with HPC system
- ❖ Data visualization with ggplot (R studio)
- ❖ Evolutionary & Phylogenetic Analysis (BEAST, Jalview, ESPript)
- ❖ Handling of antimicrobial resistance databases and (CARD, RGI-CARD, DeepARG, Resfinder)
- ❖ Pipeline: Developed Linux-based standalone version of programs
- ❖ Motif/Profile Search Tools: Well-versed with tools like MEME/MAST, InterProScan, Pfam, etc.
- ❖ Bioinformatics Tools: Sequence alignment tools, BLAST, BLAST-Clust, CD-HIT, Psi-Pred
- ❖ Database Development: Working knowledge of XAMPP (Apache, PHP).

- ❖ Web Designing: HTML, PHP
- ❖ Basic Knowledge of genomics Tools (bwa, bowtie, samtools, bamtools, bedtools, IGV, UCSC)
- ❖ Basic knowledge of various tools such as Conda, Git/GitHub

Employment/Research Experience

1. Currently working as an **Assistant Professor (Grade-I)** at the Department of Biotechnology, Jaypee Institute of Information Technology from (10th July 2025-Present)
2. **Project Scientist-I** with Dr. Dinesh Gupta, Translational Bioinformatics Group, International Centre for Genetic Engineering and Biotechnology (ICGEB) India. (16th Oct 2023 - 9th July 2025).
3. **Project Assistant** at Translational Bioinformatics Group International Centre for Genetic Engineering and Biotechnology (ICGEB) India. (28th July 2023 - 15th Oct 2023)
4. **Research Associate** at Cancer Science Institute, National University of Singapore, Singapore under the project Mutational Landscape of Microsatellite Unstable Cancers. (January 2023 - May 2023).
5. **Junior Research Fellow** at DBT-Bioinformatics Infrastructure Facility, Sri Venkateswara College, University of Delhi (August 2015- August 2016).
6. **Research Trainee** at DBT-Bioinformatics Infrastructure Facility, Sri Venkateswara College, University of Delhi (May-July 2015).

Faculty Development Program (FDP)

1. Participated in Six-Days Faculty Development Program (FDP) on “***One Health – Innovative Solutions for Health, Productivity and Conservation Challenges***”, Department of Biotechnology, JIIT Noida, July 14-19, 2025

Invited Talks/Keynote Speaker

1. 2024: Invited as a **Panelist** to an insightful session on "**Bioinsight Forum: A Panel Discussion on Drug Discovery & Bioinformatics**" at Delhi Technological University (DTU) in association with **ISCB-SC RSG-India and BioSoc-DTU** (February 09, 2024).
2. 2023: Delivered a lecture cum Hands-on session on the topic “**Hands-on Approach to Modelling and Understanding Hidden Markov Models**” at the workshop on “Artificial Intelligence in Modern Biology” (September 12-14, 2023) held at ICGEB, New Delhi, India

Internship Project (M.Tech)

Title: “Investigation of HIV-Host protein interactions in Early HIV-infection stage by Molecular Docking & Interaction Networks Studies”

Supervisor: Dr. N. Latha

Organization: DBT- Bioinformatics Infrastructure Facility, Sri Venkateswara College, University of Delhi, India 2014-2015

Honors & Awards

1. 2025: Abstract selected for poster presentation in **American Society for Microbiology (ASM) Microbiome 2025 meeting**, in Los Angeles, June 19-23, 2022. Title of the Poster: *A deep learning-based tool for the prediction of P-Glycoprotein inhibitors and substrates*
2. 2022: Received 3rd Prize (Story entitled: *Computers and artificial intelligence as weapons against antimicrobial resistance*) in Augmenting Writing Skills for Articulating Research (AWSAR) National Award conducted by **Department of Science and Technology (DST), Govt. of India**, under the Ph.D. category.
3. 2022: Abstract selected for onsite participation in **American Society for Microbiology (ASM) Microbiome 2022 meeting**, in Washington, DC, June 9-13, 2022 (Not Availled).
4. 2022: Onsite participation and poster presentation with full financial support at the **32nd European Congress of Clinical Microbiology & Infectious Diseases** in Lisbon, Portugal from April 23-26.
5. Selected for 32nd ECCMID outreach grant (**Bill & Melinda Gates Foundation**) in Lisbon, Portugal for onsite participation.
6. 2020: Selected for volunteer program in 19th International Conference on Bioinformatics, (InCoB 2020) Virtually (25-29, November 2020) for leadership and contribution to the success of InCoB 2020.
7. 2019: Selected for participation with full financial support in **Antimicrobial Resistance in Bacterial Pathogens** at Hanoi Vietnam (29 September - 4 October). Organized by **Wellcome Genome Campus Advanced Courses & Scientific Conferences (WGCAC) Cambridge**.
8. **Got Best Poster award** at National Symposium on Database Development & Biocuration (NSDDB), University of Delhi South Campus (April 6-7, 2018)
9. Organizer at 8th National Science Day Symposium, University of Delhi South Campus (27-28 February 2019)
10. 2018: **Senior Research Fellowship (Inspire Fellow)**, Department of Science & Technology, Govt. of India.
11. 2016: **Junior Research Fellowship (Inspire Fellow)**, Department of Science & Technology, Govt. of India.
12. 2015-2016: **Junior Research Fellow**, Department of Science & Technology, Govt. of India.
13. 2014-2015: **DBT scholarship** for studentship to earn M.Tech Dissertation from Bioinformatics Infrastructure Facility Sri Venkateswara College.
14. 2015: Qualified Biotech Industrial Training Program (BITP) conducted by Biotech Consortium India Limited (BCIL).
15. 2015: Secured **1st position in M.Tech (Bioinformatics) Gold Medalist**.
16. Several prizes and awards in curricular and co-curricular activities at the school and college level

Conferences/Workshop/Training/Symposia

International

1. 2023: Attended Indo-US International Conference and Workshop on Antimicrobial Resistance and Human Microbiome (ARHM) (November 15-18, 2023) held at MK Bhan Auditorium THSTI, New Delhi

2. 2020: 19th International Conference on Bioinformatics, (InCoB 2020) Virtually (25-29, November 2020) Title of the Demo-track Talk: BacEffluxPred: A two-tier system to predict and categorize bacterial efflux-mediated antibiotic resistance proteins (December 26, 2020)
3. 2020: International online conference COVID-19 Biotechnology: A way forward Manipal University Jaipur (April 30, 2020)
4. 2020: International Conference on Drug Discovery (29 February-March 2, 2020) BITS Hyderabad
5. 2020: Actively participated and completed the assignment as part of the two-session EMBO | IndiaBioscience
6. Grants Writing Workshop held from 18 June 2020 - 10 July 2020 under the aegis of EMBO | IndiaBioscience Online Seminars.
7. 2019: Programmatic access to Protein sequence, function, and structure with Uniprot and PDBe Jawaharlal University New Delhi (29 September 2019)
8. 2018: International Conference on Bioinformatics, Jawaharlal University New Delhi (26-28 September 2018) Title of the Poster: Comparative analyses of antibiotic resistance in pathogenic microbes

National

1. 2023: Attended prestigious event “Global Bio India 2023” (December 4-6, 2023) held at the Bharat Mandapam at Pragati Maidan, New Delhi, India
2. 2019: National Symposium on Database Development and Biocuration, University of Delhi South Campus (December 17-18, 2019) Title of Talk: *In-silico* Prediction tools for investigation of antimicrobial resistance genes in various –omics dataset (December 17, 2019)
3. 2019: DST INSPIRE Fellowship Review Meeting, Chandigarh University (19-21 September 2019) Title of the Poster: Comparative analyses of antibiotic resistance in pathogenic microbes (September 20, 2019)
4. 2019: 8th National Science Day Symposium, University of Delhi South Campus (27-28 February 2019) Title of the Poster: In-silico tools for comparative analyses of antibiotic resistance in pathogenic microbes (February 27, 2019)
5. 2019: National Symposium on Recent Trends, Development, and Methodologies in Biophysics organized by the Department of Biophysics University of Delhi South Campus (September 7, 2019)
6. 2015: “Symposium in Mathematical & Computational Biology”, National Network for Mathematical and Computational Biology (NNMCB), Delhi Node and Sri Venkateswara College, University of Delhi.
7. 2015: “Symposium on Proteomics”, Department of Biochemistry, Sri Venkateswara College, University of Delhi.
8. 2013: Deeksha Pandey (2013), “A review of a physiological model of heart as a specific case of arteries”, presented a poster at the 15th “Annual Conference of Society of Statistics Computer and Application” in Banasthali Vidyapith, Rajasthan.
9. 2012: 6-day workshop on “Studies in Environmental Analysis”, Department of Chemistry, Biotechnology and Microbiology Dayanand Girls P.G. College, Kanpur.
10. 2012: Deeksha Pandey (2012), “Genetically Modified Crops”, presented a poster in “Emerging frontiers and challenges” in Biotechnology, Department of Biotechnology, Dayanand Girls PG College, Kanpur.
11. 2011: (60 days B.Sc. Summer internship project) “Liver Function Test” from “PPM Medical Research & Trauma Center (P) Ltd.” Kanpur.

12. 2009: “Nucleus Biotech Cup -2009”, Nucleus Biozone Pvt. Ltd., Pioneer Biotech Company of North India in D. G. College Kanpur.

References

1. Prof. Manish Kumar (PhD Supervisor) Associate Professor & Head, Department of Biophysics, University of Delhi South Campus, New Delhi, India Phone: +91-9717914111 E. Mail: manish@south.du.ac.in	2. Prof. Manisha Goel (RAC member) Professor, Department of Biophysics University of Delhi South Campus, New Delhi, India Phone: +91-8588839751 E. Mail: manishagoel@south.du.ac.in	3. Dr. Dinesh Gupta (Current PI) Group Leader, Translational Bioinformatics Group (TBG Lab) International Centre for Genetic Engineering and Biotechnology (ICGEB), New Delhi, India Phone: +91-9312304662 E. Mail: dinesh@icgeb.res.in	4. Prof. N. Latha (M.Tech Supervisor) HoD & Professor Biotechnology School of Engineering and Applied Sciences, Bennett University E. Mail: n.latha@bennett.edu.in
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I hereby declare that all above mentioned is true to my knowledge.

DEEKSHA PANDEY